Name of Student: Rikisha Bhattacharya	**PRN: 25070122155**
Semester: III	**Year: AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No: 18	
TITLE: Java GUI Button Event Handling	

Problem Statement

Write a Java GUI program to demonstrate button click event handling.

Learning Objectives

To implement button click events using Java event handling mechanisms.

1. Create a GUI calculator where buttons perform addition and subtraction.

Programme:

Untitled-1 | Calculatorr.java

src > Calculatorr.java > ...

```java
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class Calculatorr extends JFrame implements ActionListener {

    JTextField display;

    double num1, num2, result;
    char operator;

    JButton[] numberButtons = new JButton[10];
    JButton add, subtract, multiply, divide, equals, clear;

    Calculatorr() {

        setTitle(title: "Calculator");
        setSize(width: 150, height: 350);
        setLayout(new BorderLayout());

        // Display
        display = new JTextField();
        display.setFont(new Font(name: "Arial", Font.BOLD, size: 25));
        display.setEditable(b: false);

        // Display height
        display.setPreferredSize(new Dimension(width: 0, height: 240));

        add(display, BorderLayout.NORTH);

        // Button panel
        JPanel panel = new JPanel();
        panel.setLayout(new GridLayout(rows: 4, cols: 4, hgap: 5, vgap: 5));

        // Number buttons
        for (int i = 0; i <= 9; i++) {

            numberButtons[i] = new JButton(String.valueOf(i));

            // Number button font size
            numberButtons[i].setFont(
                new Font(name: "Arial", Font.BOLD, size: 25)
            );

            numberButtons[i].addActionListener(this);
        }

        // Operation buttons
        add = new JButton(text: "+");
        subtract = new JButton(text: "-");
        multiply = new JButton(text: "*");
        divide = new JButton(text: "÷");
        equals = new JButton(text: "=");
        clear = new JButton(text: "C");

        // Make operation buttons same size as number buttons
        add.setFont(new Font(name: "Arial", Font.BOLD, size: 25));
        subtract.setFont(new Font(name: "Arial", Font.BOLD, size: 25));
        multiply.setFont(new Font(name: "Arial", Font.BOLD, size: 25));
        divide.setFont(new Font(name: "Arial", Font.BOLD, size: 25));
        equals.setFont(new Font(name: "Arial", Font.BOLD, size: 25));
        clear.setFont(new Font(name: "Arial", Font.BOLD, size: 25));

        // Action listeners
        add.addActionListener(this);
        subtract.addActionListener(this);
        multiply.addActionListener(this);
        divide.addActionListener(this);
        equals.addActionListener(this);
        clear.addActionListener(this);

        // Arrange buttons

        panel.add(numberButtons[7]);
        panel.add(numberButtons[8]);
        panel.add(numberButtons[9]);
        panel.add(add);

        panel.add(numberButtons[4]);
        panel.add(numberButtons[5]);
        panel.add(numberButtons[6]);
        panel.add(subtract);

        panel.add(numberButtons[1]);
        panel.add(numberButtons[2]);
        panel.add(numberButtons[3]);
        panel.add(multiply);

```

```
        panel.add(numberButtons[0]);
        panel.add(clear);
        panel.add(divide);
        panel.add(equals);

        add(panel, BorderLayout.CENTER);

        // Full screen
        setExtendedState(JFrame.MAXIMIZED_BOTH);

        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setVisible(b: true);
    }

    public void actionPerformed(ActionEvent e) {

        // Number buttons
        for (int i = 0; i <= 9; i++) {

            if (e.getSource() == numberButtons[i]) {

                display.setText(
                    display.getText() + i
                );
            }
        }

        // Addition
        if (e.getSource() == add) {

            num1 = Double.parseDouble(
                display.getText()
            );

            operator = '+';

            display.setText(t: "");
        }

        // Subtraction
        if (e.getSource() == subtract) {

            num1 = Double.parseDouble(
                display.getText()
            );

            operator = '-';

            display.setText(t: "");
        }

        // Multiplication
        if (e.getSource() == multiply) {

            num1 = Double.parseDouble(
                display.getText()
            );

            operator = '*';

            display.setText(t: "");
        }

        // Division
        if (e.getSource() == divide) {

            num1 = Double.parseDouble(
                display.getText()
            );

            operator = '/';

            display.setText(t: "");
        }

        // Equals
        if (e.getSource() == equals) {

            num2 = Double.parseDouble(
                display.getText()
            );

            if (operator == '+') {

                result = num1 + num2;
            }

```

```
            if (operator == '-') {

                result = num1 - num2;
            }

            if (operator == '*') {

                result = num1 * num2;
            }

            if (operator == '/') {

                if (num2 != 0) {
                    result = num1 / num2;
                }
                else {
                    display.setText(t: "Cannot divide by zero");
                    return;
                }
            }

            display.setText(
                String.valueOf(result)
            );
        }

        // Clear
        if (e.getSource() == clear) {

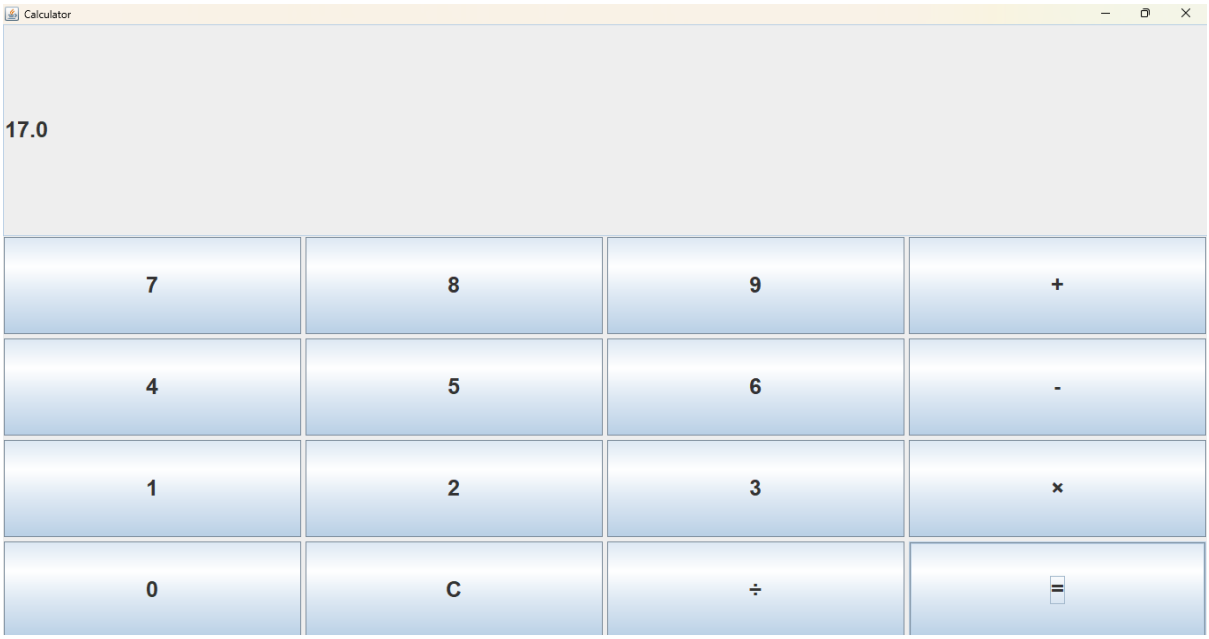
            display.setText(t: "");

            num1 = 0;
            num2 = 0;
            result = 0;
            operator = '\0';
        }
    }

    Run | Debug
    public static void main(String[] args) {

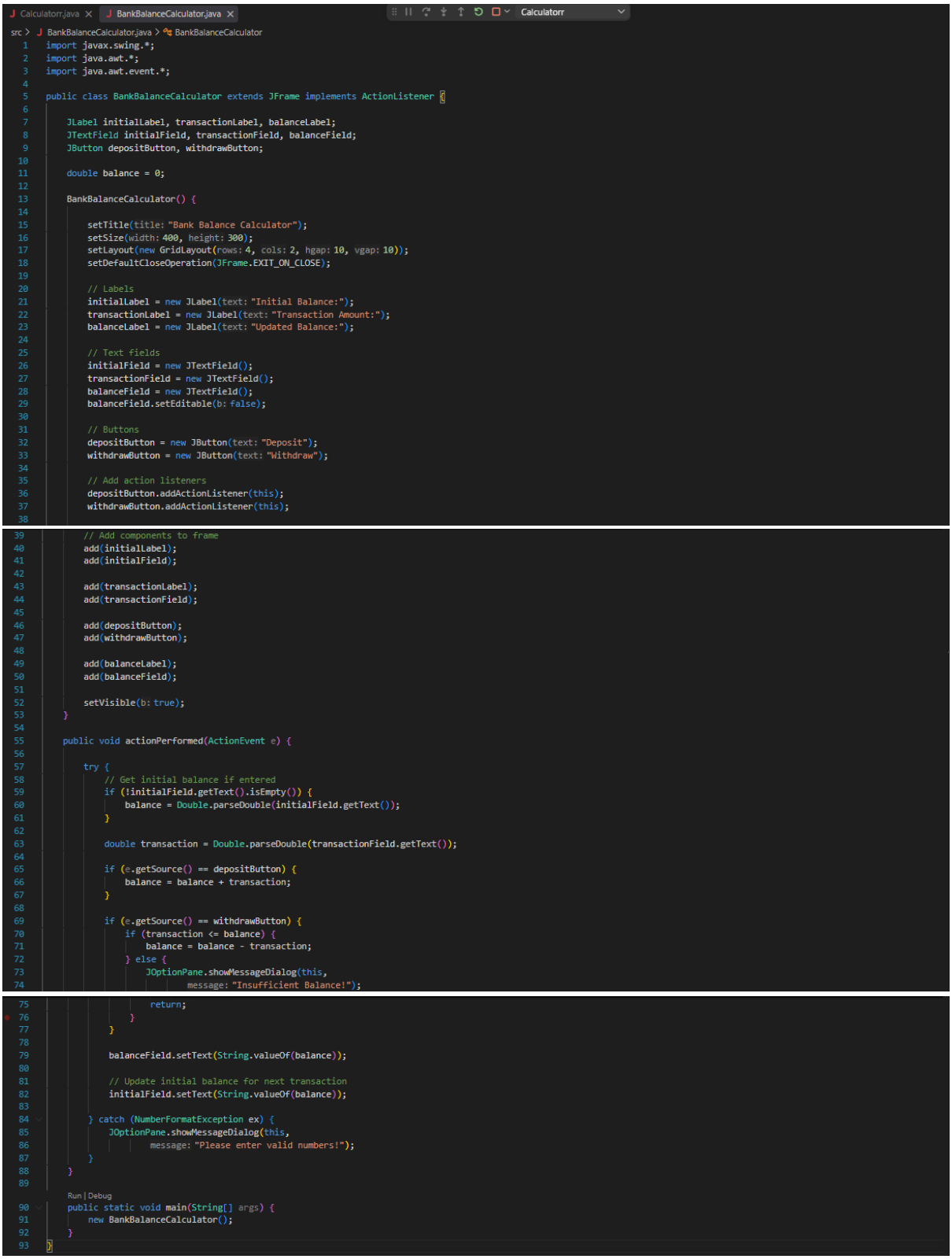
        new Calculatorr();
    }
}
```

Output:



2. Create a Bank Balance Calculator GUI application using Java Swing where the user enters the initial balance and transaction amount. Use buttons to perform addition (deposit) and subtraction (withdrawal) and display the updated balance.

Programme:



```java
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class BankBalanceCalculator extends JFrame implements ActionListener {

    JLabel initialLabel, transactionLabel, balanceLabel;
    JTextField initialField, transactionField, balanceField;
    JButton depositButton, withdrawButton;

    double balance = 0;

    BankBalanceCalculator() {

        setTitle(title: "Bank Balance Calculator");
        setSize(width: 400, height: 300);
        setLayout(new GridLayout(rows: 4, cols: 2, hgap: 10, vgap: 10));
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        // Labels
        initialLabel = new JLabel(text: "Initial Balance:");
        transactionLabel = new JLabel(text: "Transaction Amount:");
        balanceLabel = new JLabel(text: "Updated Balance:");

        // Text fields
        initialField = new JTextField();
        transactionField = new JTextField();
        balanceField = new JTextField();
        balanceField.setEditable(b: false);

        // Buttons
        depositButton = new JButton(text: "Deposit");
        withdrawButton = new JButton(text: "Withdraw");

        // Add action listeners
        depositButton.addActionListener(this);
        withdrawButton.addActionListener(this);

        // Add components to frame
        add(initialLabel);
        add(initialField);

        add(transactionLabel);
        add(transactionField);

        add(depositButton);
        add(withdrawButton);

        add(balanceLabel);
        add(balanceField);

        setVisible(b: true);
    }

    public void actionPerformed(ActionEvent e) {

        try {
            // Get initial balance if entered
            if (!initialField.getText().isEmpty()) {
                balance = Double.parseDouble(initialField.getText());
            }

            double transaction = Double.parseDouble(transactionField.getText());

            if (e.getSource() == depositButton) {
                balance = balance + transaction;
            }

            if (e.getSource() == withdrawButton) {
                if (transaction <= balance) {
                    balance = balance - transaction;
                } else {
                    JOptionPane.showMessageDialog(this,
                            message: "Insufficient Balance!");
                    return;
                }
            }

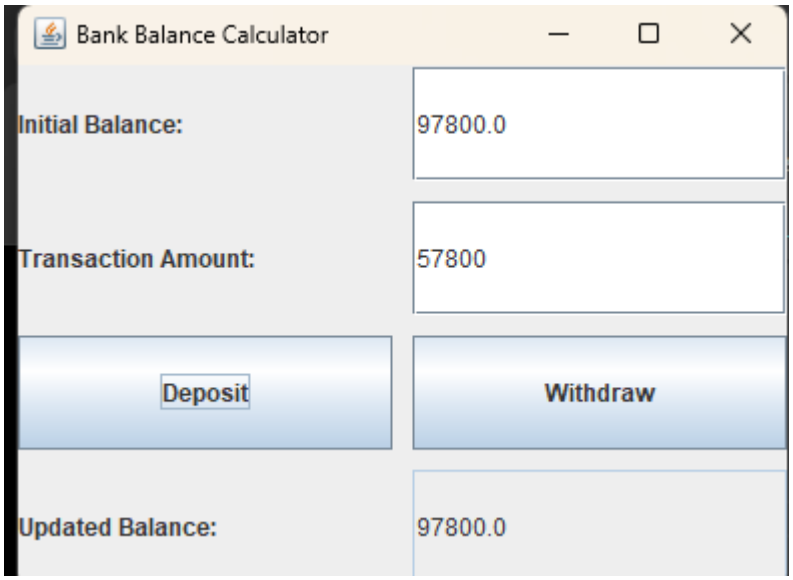
            balanceField.setText(String.valueOf(balance));

            // Update initial balance for next transaction
            initialField.setText(String.valueOf(balance));

        } catch (NumberFormatException ex) {
            JOptionPane.showMessageDialog(this,
                    message: "Please enter valid numbers!");
        }
    }

    Run | Debug
    public static void main(String[] args) {
        new BankBalanceCalculator();
    }
}
```

Output:



Conclusion:

Thus, the Java GUI programs were successfully implemented using Java Swing and event handling mechanisms. The button click events were used to perform addition and subtraction operations in the calculator and to update the bank balance for deposits and withdrawals. This experiment helped in understanding GUI components, event listeners, and user interaction in Java.